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Submitter Information

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General Comment

See attached file(s)

Attachments

OSTP AI RFI Action Plan Response

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Executive Office of the President
2415 Eisenhower Avenue
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Submitted by email to [redacted]

Re: Request for Information (RFI) on the Development of an Artificial Intelligence (AI) Action Plan (“Plan”)

Executive Summary

The NSF AI Institute for Research in Trustworthy AI in Weather, Climate, and Coastal Oceanography (AI2ES) strongly supports the OSTP’s efforts to outline policy and enact action to enhance America’s AI capabilities and innovations. AI is growing and expanding exponentially and it is important that the US continue to lead in this important area.

Our recommendations fall into the following thematic areas:

1. The need for expanded federal investments in use-inspired AI
2. The need for new approaches to cross-sector partnerships for AI acceleration
3. The need for expanded investment into AI workforce development and education.

About NSF AI2ES

AI2ES is one of the inaugural NSF AI institutes, funded in 2020. As an AI institute, our primary focus is to develop novel AI focused on real-world impact. AI2ES is focused on developing trustworthy AI for a variety of Earth system science applications. We develop AI methods for predicting and understanding extreme weather and ocean hazards that threaten U.S. lives and property. As a leader in this space, we directly partner with private industry companies, federally funded research laboratories, and key public and private sector decision makers whose decisions rely on weather predictions and affect society at large, to develop impactful products and systems for contexts where AI transparency and assurance is most critical. Our research spans the full cycle of innovation from fundamental AI research to operational deployment enabling the rapid integration of beneficial and trusted AI models for increased societal impact.

Need for expanded federal investments in use-inspired AI

NSF AI2ES encourages expanded federal investments in use-inspired AI. By focusing on problems of high relevance to society, this will both accelerate the development of novel AI methods and directly address critical needs of society.

AI2ES has focused on the prediction of extreme weather and ocean hazards, which has far reaching impacts on American infrastructure and business. Extreme weather events such as tropical cyclones and coastal inundation threaten the communities, industry, and military operations along our coastlines. Winter weather including snow blizzards and icy conditions impede the reliability and efficiency of transportation and shipping across the nation. Severe storm hazards like hail, tornadoes, and damaging winds produce billions of dollars of damage to privately owned and publicly maintained infrastructure. AI models that predict and improve the understanding of these hazardous events must be explainable, understandable, and reliable to the professional end-users who depend on them. Other AI institutes have focused on other societally pressing needs, including AI for education and health. These needs continue to grow; additional investment in use-inspired AI will strengthen American leadership in AI and benefit our nation financially.

This need for stakeholder-informed AI is not exclusive to the weather and ocean communities. As the AI expands and becomes ubiquitous in more industrial sectors, the need for AI assurance and trust is growing. **We recommend that as a nation we stay ahead of the curve by making strategic federal investments, specifically by increasing funding for user-informed and use-case inspired AI.** We encourage American investments to evaluate AI safety and risks, enhance AI explainability, accountability, and assurance, and to ensure maximal impact through the inclusion of these cross-sectorial partnerships. This includes the co-development of AI systems with private industry, academia, public sector, and non-profits.

Cross-sector partnerships for AI

NSF AI2ES encourages the creation of novel funded avenues to build cross-sector partnerships across private industry, academia, and federal institutions.

Our ability to produce trusted and assured AI models is dependent on our high-quality relationships with our industry and government partners. We prioritize the co-development of AI models with the professional end-users of this technology, ensuring AI model safety and minimizing unaddressed risk factors. Through this collaborative approach we speed up and improve the AI modeling R&D process. Compounding the efficiency gains, we have found that it is necessary to engage in cross-sectoral partnership and integrate stakeholder feedback to produce societally and industrially impactful AI. AI models are among–if not the–most powerful technologies available to improve the ability of human decision makers to incorporate past and predictive information into their decisions. Investing more in learning how human decision makers and models can best work together will further amplify efficiency gains.

Creating partnerships across sectors requires significant investment of time and resources on all sides. Federal investment and new pathways for partnerships could significantly enhance such collaborations. This would enhance both the development of fundamental new methods and the deployment of such methods.

Expanded workforce development and educational investment in AI

NSF AI2ES recommends increased investment in AI education at the community college and undergraduate level to maximize the impact of the emerging workforce. We also recommend federal investments in interdisciplinary research to bridge the knowledge gap between AI and other domains.

As AI development and use accelerates rapidly, the United States requires an increasing number of qualified, well-trained, and technologically advanced professionals to sustain American AI dominance. From early education to workforce retraining, we must ensure every level of the American workforce has appropriate resources and educational material to learn how to utilize or create AI systems. This will require federal investment in education at all levels, from K-12 to community colleges and universities.

There is a growing need for AI trained professionals across all industrial sectors, and expanding AI education alone will not satisfy this increasing demand. Increased federal investments in interdisciplinary research can provide students with opportunities to solve real-world problems with AI and contribute to foundational knowledge on how to create AI for specific use cases. This will further improve the technological sophistication of the US workforce.

Conclusion

As AI is expanding to new industrial sectors and establishing itself as the new frontier of innovation, it is critical that the American vision for progress addresses the new opportunities and threats that come with new industry-disrupting advancements. In our submission, we advocate for federal support for use-inspired AI research, cross-sector collaborations and partnerships, and workforce development at the college and university level. Supported by our extensive experience in AI for real-world impact, we believe federal support of these foci is pivotal for securing future American success, both in the short-term and long-term, ensuring our nation benefits scientifically, technically, and financially in this rapidly changing landscape.